Program: Building a basic webapp using maven

Step 1: Build the project

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp001 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 10 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/MyMavenWebApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.361 s
[INFO] Finished at: 2026-04-08T08:27:59+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops$
```

Step 2: Check project strucutre

run tree

```
harith@harith-VirtualBox:~/devops/MyMavenWebApp001$ tree
.
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

5 directories, 3 files
harith@harith-VirtualBox:~/devops/MyMavenWebApp001$
```

Step 3: Edit pom.xml , add following dependecies and plugins

```
<!-- Servlet API for Java EE -->
    <dependency>
      <groupId>javax.servlet</groupId>
      <artifactId>javax.servlet-api</artifactId>
      <version>4.0.1</version>
      <scope>provided</scope>
    </dependency>


 <build>
```

```
  <plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version> <!-- Update to latest stable version -->
</plugin>
</plugins>
    <finalName>MyMavenWebApp</finalName>
  </build>
```

Step 4: run mvn clean install

```
harith@harith-VirtualBox:~/devops/MyMavenWebApp001$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenWebApp001 >-------------------
[INFO] Building MyMavenWebApp001 Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.pom (8.4 kB at 26 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.4.0/maven-war-plugin-3.4.0.jar (83 kB at 574 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servlet-api/4.0.1/javax.servlet-api-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servlet-api/4.0.1/javax.servlet-api-4.0.1.pom (16 kB at 170 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/jvnet-parent-3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/jvnet-parent-3.pom (4.8 kB at 53 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servlet-api/4.0.1/javax.servlet-api-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servlet-api/4.0.1/javax.servlet-api-4.0.1.jar (96 kB at 729 kB/s)
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenWebApp001 ---
[INFO] Deleting /home/harith/devops/MyMavenWebApp001/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenWebApp001 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenWebApp001 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenWebApp001 ---
```

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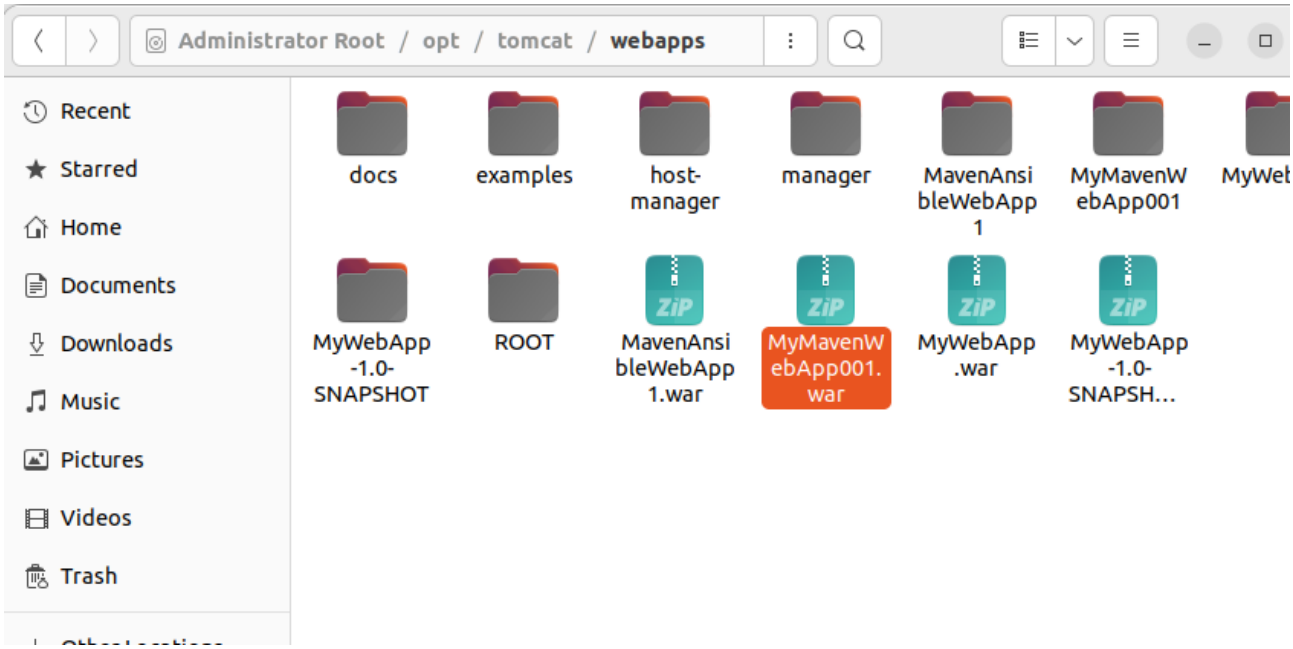
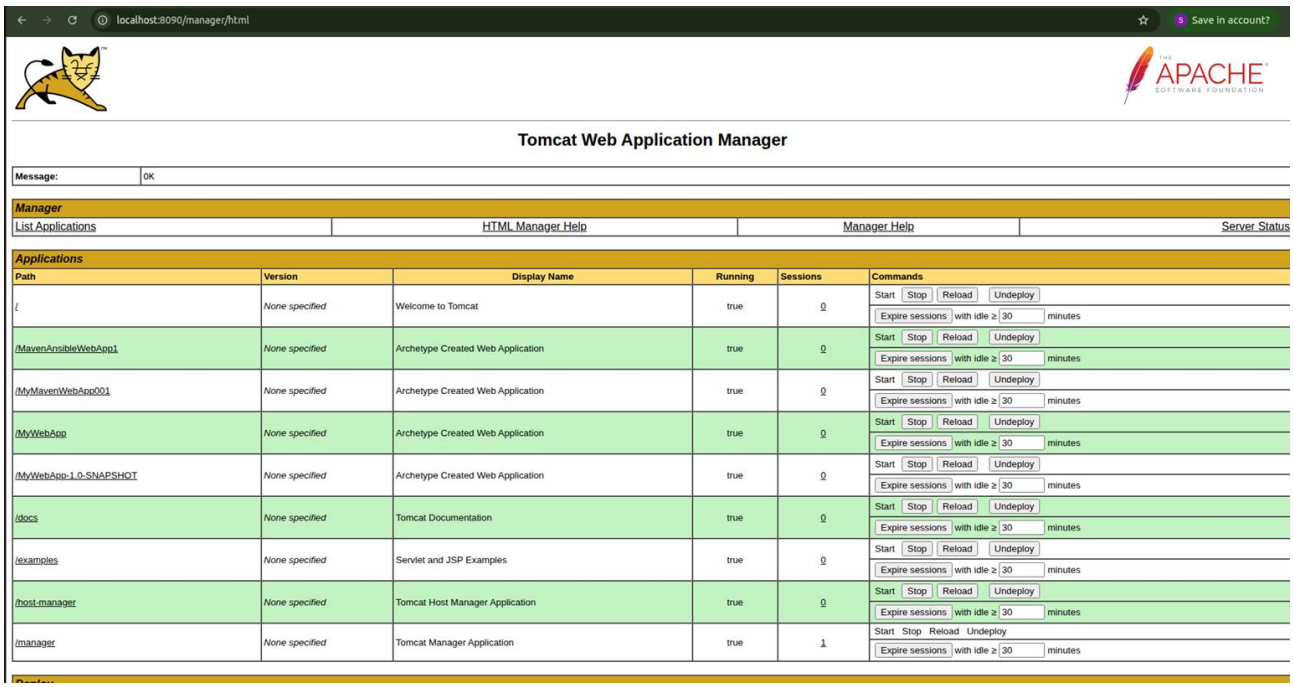
Tomcat:

- Here we manually copy the war file into opt/tomcat/webapps



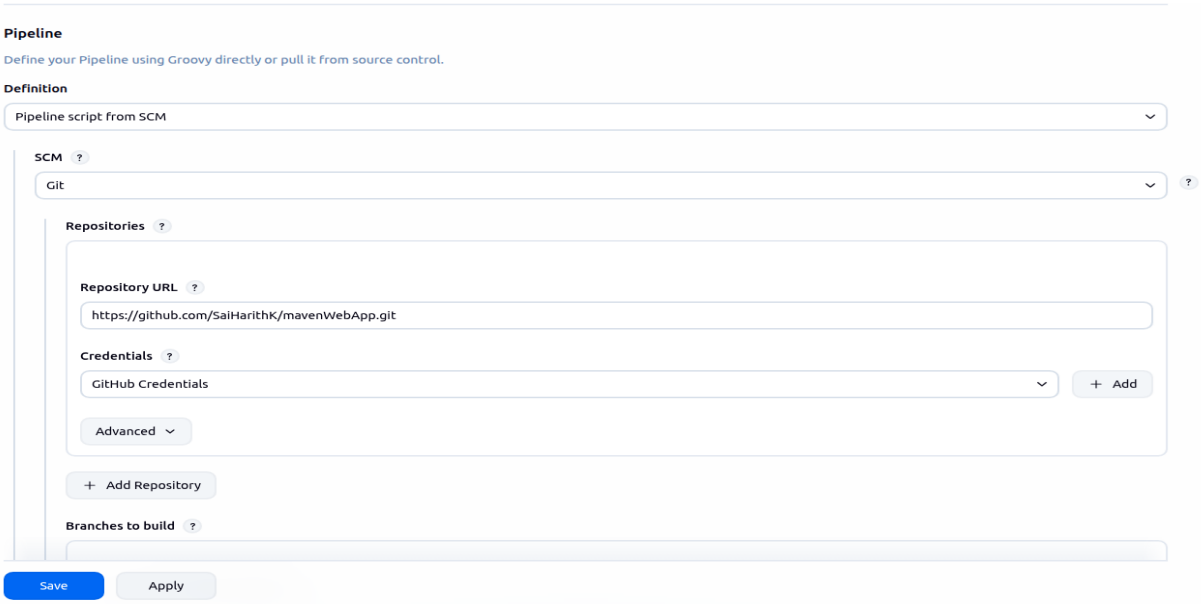
- Once its moved into webapps, we open tomact on locathost:8090



-Provide crednetials to login and click on the project name to get the output.

Step 9: Pipeline

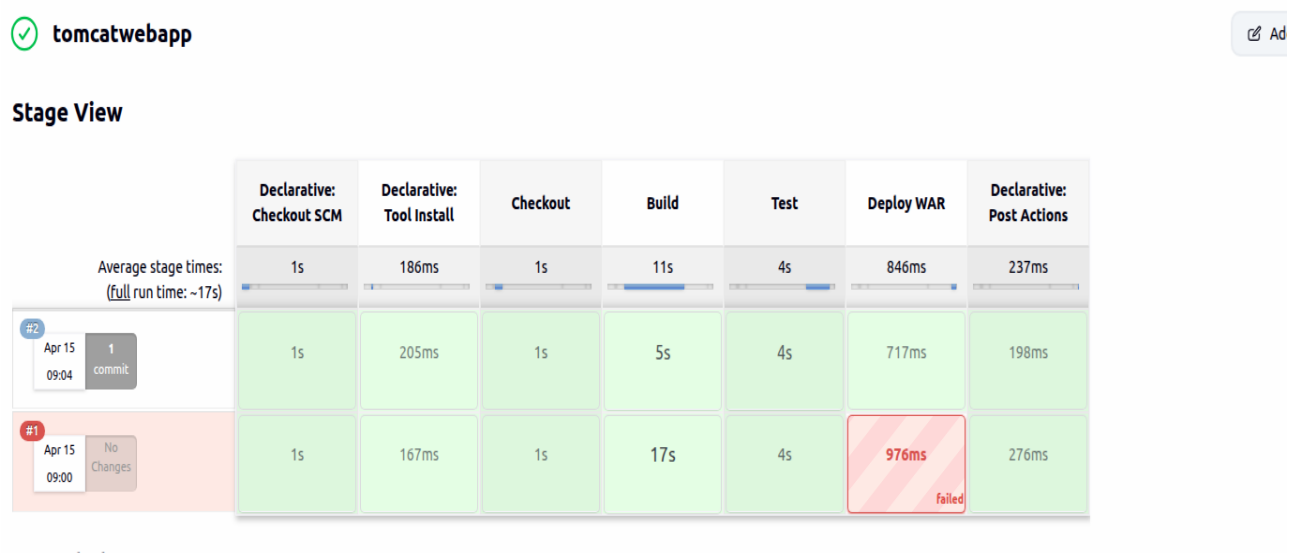
-login to jenkins account -> create a new item and configure the following settings.



Step 10: Edit Jenkins file

```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/SaiHarithK/mavenWebApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }tage('Deploy WAR') {
            steps {
                sh 'cp target/MyMavenWebApp001.war /opt/tomcat/webapps/'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
```

```
            echo 'Build failed!'
        }
    }
}
```

Step 11: Build

Click on build now and check if its successfull